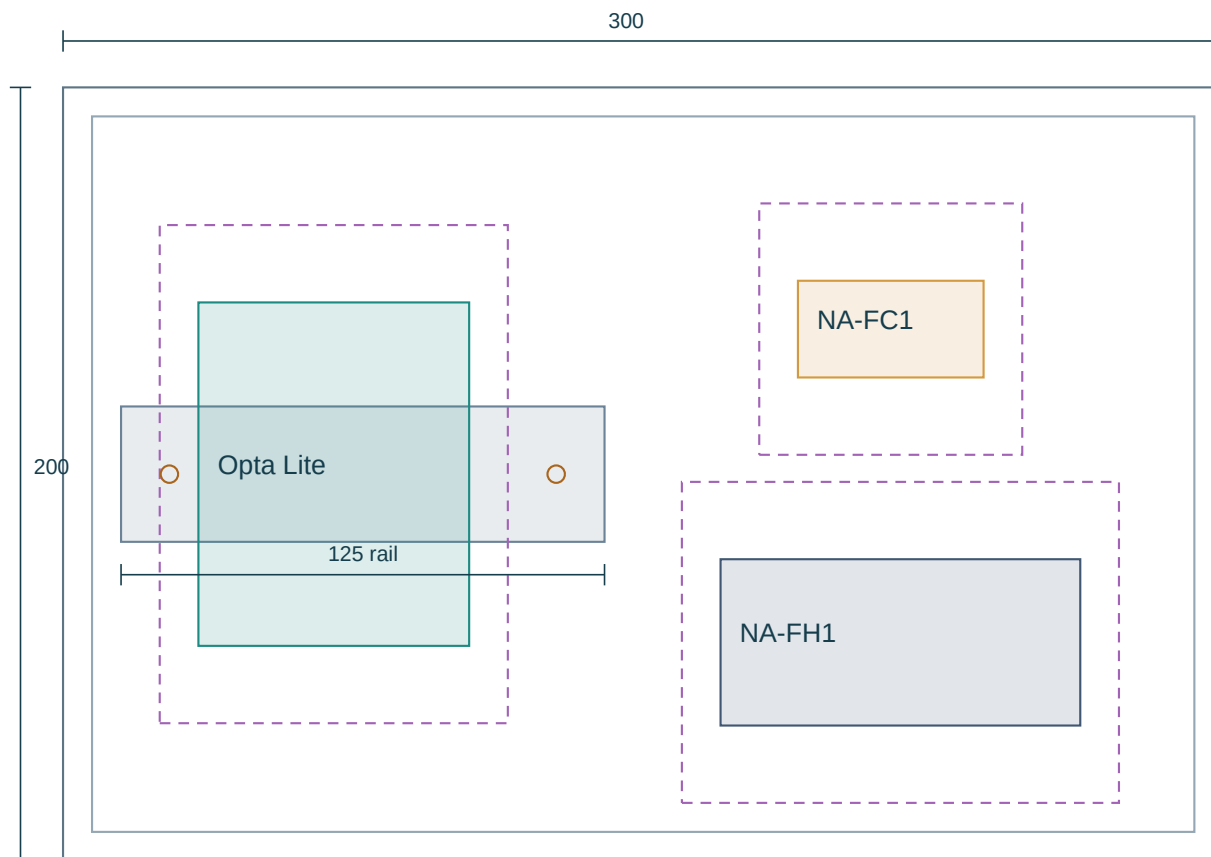


External control box: dimensioned body layout



Verified sources: Hammond1554XA2GY box; matching1554XPL panel. Opta70 x88.8 x61.1 depth reservation; hub93 x43 x12.5; speed controller48 x25 x21 in the proposed orientation. Components are simplified bodies, not exact connector models.

Checked: all3 body envelopes and3 assumed service volumes clear24 solids in the private OEM STEP, excluding intended panel support. Bodies have65 /85 /47 mm pairwise minimum separation. No physical fit result.

Design choice: retain existing493-object tower unchanged. Box is a separate, supervised-indoor module; its placement, retention and connection to the tower are not approved. Power adapter stays external. No mains inside this proposed box.

Not yet fitted: PR1, fuses/holders, terminals, cable glands, buttons and their backs, supports, functional earth and complete wiring. A body-fit pass is NOT a complete control-panel fit pass.

Datum: enclosure centre; +X right, +Y up; dimensions mm

Purple dashed = ASSUMED service/routing space, not cable specifications

Body lower-left X/Y coordinates: Opta -115 /-44.4; hub20 /-65; speed controller40 /25. DIN envelope lower-left -135 /-17.5. Each body dimension and Z coordinate is in PLACEMENT.csv. Coordinates locate proposed envelopes, NOT mounting holes or connectors.

Mounting and depth: explicit next interface

Layer / proposal	Dimension or coordinate	Evidence / release status
OEM panel top	STEP Z = -58.3744 mm	OEM panel thickness1.6256 inSTEP; nominal page rounds to2mm
DIN support envelope	35 mm wide /7.5 mm high /125 mm proposed cut	Phoenix0801733; actual profile and clips not drawn
Opta back / front	7.5 /68.6 mm above panel top	Conservative ASSUMED back plane at rail crest; actual seating unknown
Hub / controller carrier	5 mm ASSUMED lift above panel	No actual carrier, clamp, screw or magnetic-only retention approval
Proposed rail holes	X -122.5 and -22.5; Y0; diameter4.5 mm	100mm centres; aligned-to-slot cut ASSUMED; NOT drilling release
Panel mounting	Use1554XPL matching panel / OEM insert pattern	Do not drill plastic box or transfer part-label locations

Fit limitation: the supplied STEP contains moulded geometry, screws, gasket and a shaped panel. Published CAD intentionally uses an original simplified rectangular panel and shell. It is NOT a substitute for Hammond drawings. Panel contour, existing holes, threads, DIN clip, terminal access and button/connector backs are not manufactured details.

Assembly review sequence: inspect exact delivered box/panel/controller; establish actual panel thickness and DIN seating; fix reviewed rail/supports to panel; mechanically retain hub/controller; fit selected protection and terminations; review external source and all entries; check lid/clip/connector access, strain relief, clearances and heat. No energized wiring steps are supplied.

Service decision still open: NA-FC1 is inside the box. Its dial and switch are not accessible through a closed lid in this layout. Do not operate with the box open just to reach it. A reviewed access strategy or different control component is needed before use. No arbitrary lid cutouts or thermal vents.

Release holds: protective PR1 and fault coordination, real wires/contact gauges, retention and mounting strength, connector bends, heat rise, enclosed controls/access, functional earth, whole-device ingress and placement. The stock IP68 rating does not apply to the completed or drilled module. Physical tests0. All relay commands remain disabled in default firmware.

Sources: exact URLs and OEM STEP hash are in FIT_CHECKS.json. Source PDFs/STEP stay private local references; redistribution permission was not established. Editable FCStd/STEP here contain only original envelope geometry. No purchase, supplier contact, new software installation or tower geometry change.